TOBB ETÜ

University of Economics & Technology

Proposal

Final Stack

Ali Belikırık
Beyza Değirmenci
İnan Özşahin
Özge Doğan

1. Project Title: Portfolyo Bilgi Sistemi (PBS)

2. Problem Definition & Objectives

- **Problem Definition (1–2 paragraphs):**

  In today's education system, tracking student development and learning processes is not limited to exam grades alone. The projects, assignments, artwork, and other products that students create are much more effective in demonstrating their real abilities and progress. However, there are significant challenges in systematically collecting, organizing, and sharing these works with parents.

  Traditional physical portfolios are bulky, difficult to transport, and carry the risk of deterioration over time. Teachers spend considerable time categorizing student work, linking it to learning objectives, and presenting it meaningfully to parents. Parents, on the other hand, can only view their children's school work during parent-teacher conferences or report card periods, making continuous monitoring impossible. Additionally, there are no suitable tools for students to actively participate in their own learning processes and develop self-assessment skills. This project aims to provide a digital solution to all these problems.

- **Objectives:**
  - To enable students to store and organize their learning products digitally.
  - To allow teachers to evaluate and tag student works according to PYP (Primary Years Programme) learner profile attributes.
  - To provide parents with easy access to selected student works.
  - To create a visual tracking system showing the student’s progress over time.
  - To develop a secure, user-friendly, and accessible digital platform.

3. Scope

- A web-based digital portfolio system will be developed.
- User roles: student, teacher, parent, and administrator.
  - Students will be able to upload their works (text, image, video, etc.).

    - Teachers will categorize and tag works based on PYP learner profile attributes.
    - Parents will access selected works and provide feedback.
    - Administrators will manage user accounts and system settings.
- The system will be designed to be mobile-friendly and data-secure.

4. User Profile

- **Target Users:**
    - **Students:** Upload and manage their own works to document learning progress.
    - **Teachers:** Evaluate, tag, and provide feedback on student works.
    - **Parents:** View selected student works and monitor their child's progress.
    - **Administrators:** Manage system users, permissions, and overall configuration**.**

5. Anticipated Challenges & Constraints

Technical Hurdles:

- Secure upload and storage of multimedia files.
- Designing a flexible and dynamic tagging system.
- Developing an intuitive user interface suitable for different age groups.

Time/Resources Limits:

- A small development team may restrict task distribution.
- Possible hosting and cloud storage costs.

Ethical or Regulatory Considerations:

- Managing parental access permissions carefully.
- Ensuring child-appropriate content and safe data handling.

6. Data Sources (if any)

User-Generated Content:

- Student work (images, photos, videos, PDF, document files)
- Student self-assessment notes and descriptions
- Teacher feedback, comments, and assessments
- Skill and competency tags

System Data:

- User profile information (name, surname, role, class, school)
- File metadata (upload date, size, format, status)
- Tagging and categorization data
- Activity logs and usage statistics
- Notification and messaging records

Reference Data:

- List of basic skills and competency areas (communication, creativity, problem-solving, etc.)
- Age-appropriate development indicators
- Sample assessment criteria and rubrics

Potential External Sources:

- Consultation from education experts
- Feedback and suggestions collected from pilot school implementations
- User experience research from similar systems